



Choose the right tool for the job

As a data analyst, you will usually have to decide which program or solution is right for the particular project you are working on. In this reading, you will learn more about how to choose which tool you need and when.

Depending on which phase of the data analysis process you're in, you will need to use different tools. For example, if you are focusing on creating complex and eye-catching visualizations, then the visualization tools we discussed earlier are the best choice. But if you are focusing on organizing, cleaning, and analyzing data, then you will probably be choosing between spreadsheets and databases using queries. Spreadsheets and databases both offer ways to store, manage, and use data. The basic content for both tools are sets of values. Yet, there are some key differences, too:

Spreadsheets

Accessed through a software application

Structured data in a row and column format

Organizes information in cells

Provides access to a limited amount of data

Manual data entry

Generally, one user at a time

Controlled by the user

Databases

Database accessed using a query language

Structured data using rules and relationships

Organizes information in complex collections

Provides access to huge amounts of data

Strict and consistent data entry

Multiple users

Controlled by a database management system

You don't have to choose one or the other because each serves its own purpose. Generally, data analysts work with a combination of the two, as both tools are very useful in data analytics. For example, you can store data in a database, then export it to a spreadsheet for analysis. Or, if you are collecting information in a spreadsheet, and it becomes too much for that particular platform, you can import it into a database. And, later in this course, you will learn about programming languages like R that give you even greater control of your data, its analysis, and the visualizations you create.

Key takeaways

The choice of data analysis tools depends on the specific task at hand. Spreadsheets are suitable for organizing, cleaning, and analyzing small to medium datasets. Databases are ideal for storing, managing, and analyzing large and complex datasets. Data analysts often use a combination of spreadsheets, databases, and programming languages to effectively handle a wide range of data analysis tasks.